Koa Shen

Mechanical Engineer

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EDUCATION

University of California, Santa Barbara — Expected June 2028
B.S. Mechanical Engineering | Major GPA: 3.97 | Cumulative GPA: 3.94 — Santa Barbara, CA

- Dean's Honors every quarter. Coursework: Dynamics, Mechanics of Materials, Circuits, MATLAB. Current: Dynamical Systems and Machine Learning.

EXPERIENCE

Mechanical Engineering Intern — Summer 2026 - Present
SLAC National Accelerator Laboratory, LCLS — Menlo Park, CA (part-time remote)

- Built an open-source Python/Drake digital twin for a 30-DOF, 305-body beamline assembly; millisecond collision checks exposed at least five pinch points.
- Developed 15 reusable motion-stage models now being adopted across three LCLS assemblies; applied simulation findings to optics packaging and alignment redesigns.
- Designed mounts and detector strain relief, evaluated stage and mount load cases for $100k+ detectors, and released Solid Edge drawings with GD&T through Teamcenter PDM.

Fabrication Team Lead — Fall 2024 - Present
Gaucho Racing, Formula SAE Electric | UC Santa Barbara — Santa Barbara, CA

- Led 15 teammates through GR26 driver-safety-system design, fabrication, and installation; the package passed Formula SAE technical inspection.
- CNC-machined 33 suspension components from 4130 steel and 7075-T6 aluminum; held clevis and rod-end bores within a 0.0004-in tolerance window and bearing interfaces within 0.0005 in.
- Worked with designers on GD&T and manufacturability; standardized five spindle interfaces to reuse proven CAM programs and reduce setup and programming work.

Undergraduate Researcher — Fall 2025 - Present
Hawkes Lab, UC Santa Barbara — Santa Barbara, CA

- Designed and 3D-printed custom molds with complex curvature for composite parts; fabricated parts by wet layup and prepreg across different weaves and fabrics.
- Experimented with vacuum-bagging setups and iterated on mold design to improve part stiffness.

Maintenance Technician — Summer 2025
Exploratorium — San Francisco, CA

- Serviced 100+ interactive exhibits and led three major overhauls through failure diagnosis, reverse engineering, manual machining, and welding.
- Redesigned a load-spreading bezel to address acrylic cracking and avoid $3,000+ in replacement costs; documented CAD and remanufacturing instructions for future technicians.

SELECTED PROJECTS

Compact Cycloidal Reducer & Stepper Package — Summer 2026 - Present

- Designed a 15:1 cycloidal reducer within a NEMA 17 motor's 42 mm square footprint; fabricated components and assembled the stepper, driver, and encoder package.
- Built PlatformIO C++ firmware with AS5600 encoder feedback; bench-validated single-joint position correction and missed-step detection.

FSAE Chassis Welding Jig Evolution — Winter 2024 - Spring 2026

- Designed a modular GR25 welding fixture that located 81 chassis tubes within 0.050 in of CAD node positions and reduced assembly time 50% versus the prior fixture.
- Developed a GR26 hybrid fixture with 3D-printed locating brackets and reused 80/20 and plywood, reducing fixture assembly time a further 30%.

TECHNICAL SKILLS

Mechanical Design & Analysis: SolidWorks, Solid Edge, Teamcenter PDM, GD&T, DFM, FEA, Statics/dynamics
Manufacturing & Materials: CNC milling and turning, Mastercam, Manual machining, Welding, Composite layup, 4130 steel, 7075-T6 aluminum, 3D printing
Software & Controls: Python, C/C++, MATLAB, Drake kinematics, PlatformIO, Stepper motors, AS5600 encoders, Git